

CHAPTER 1

OPTIONS

This chapter is focused on real life uses of options for directional investors, for example using options to replace a long position in the underlying, to enhance the yield of a position through call overwriting, or to provide protection from declines. In addition to explaining these strategies, a methodology to choose an appropriate strike and expiry is shown. Answers to the most common questions are given, such as when an investor should convert an option before maturity, and the difference between delta and the probability that an option expires in the money.

1.1: OPTION BASICS

This section introduces options and the history of options trading. The definition of call and put options, and how they can be used to gain long or short equity exposure, is explained. Key definitions and terminology are given, including strike, expiry, intrinsic value, time value, ATM, OTM and ITM.

HISTORY OF VOLATILITY TRADING

While standardised exchange traded options only started trading in 1973 when the CBOE (Chicago Board Options Exchange) opened, options were first traded in London from 1690. Pricing was made easier by the Black-Scholes-Merton formula (usually shortened to Black-Scholes), which was invented in 1970 by Fischer Black, Myron Scholes and Robert Merton.

Option trading exploded in the 1990s

The derivatives explosion in the 1990s was partly due to the increasing popularity of hedge funds, which led to volatility becoming an asset class in its own right. New volatility products such as volatility swaps and variance swaps were created, and a decade later futures on volatility indices gave investors listed instruments to trade volatility. In this chapter we shall concentrate on option trading.

CALL OPTIONS GIVE RIGHT TO BUY, PUTS RIGHT TO SELL

A European call is a contract that gives the investor the right (but not the obligation) to buy a security at a certain strike price on a certain expiry date (American options can be exercised before expiry). A put is identical except it is the right to sell the security.

Call option gives long exposure, put options give short exposure

A call option profits when markets rise (as exercising the call means the investor can buy the underlying security cheaper than it is trading, and then sell it at a profit). A put option profits when markets fall (as you can buy the underlying security for less, exercise the put and sell the security for a profit). Options therefore allow investors to put on long (profit when prices rise) or short (profit when prices fall) strategies.

SELLING OPTIONS GIVES OPPOSITE EXPOSURE

As a call option gives long exposure to the underlying security, selling a call option results in short exposure to the underlying security. Similarly while a put option is a bearish (profits from decline in the underlying) strategy, selling a put option is a bullish strategy (profits from a rise in the underlying). While the direction of the underlying is the primary driver of profits and losses from buying or selling options, the volatility of the underlying is also a driver.

OPTIONS TRADING GIVES VOLATILITY EXPOSURE

If the volatility of an underlying is zero, then the price will not move and an option's payout is equal to the intrinsic value. Intrinsic value is the greater of zero and the 'spot – strike price' for a call and is the greater of zero and 'strike price – spot' for a put. Assuming that stock prices can move, the value of a call and put will be greater than intrinsic due to the time value (price of option = intrinsic value + time value). If an option strike is equal to spot (or is the nearest listed strike to spot) it is called at-the-money (ATM).

As volatility increases so does the price of call and put options

If volatility is zero, an ATM option has a price of zero (as intrinsic is zero). However, if we assume a stock is €50 and has a 50% chance of falling to €40 and 50% chance of rising to €60, it has a volatility above zero. In this example, an ATM call option with strike €50 has a 50% chance of making €10 (if the price rises to €60 the call can be exercised to buy the stock at €50, which can be sold for €10 profit). The fair value of the ATM option is therefore €5 ($50\% \times €10$); hence, as volatility rises the value of a call rises (a similar argument can be used for puts).

Options have greatest time value when strike is similar to spot (i.e. ATM)

An ATM option has the greatest time value (the amount the option price is above the intrinsic value). This can be seen in the same example by looking at an out-the-money (OTM) call option of strike €60 (an OTM option has strike far away from spot and zero intrinsic value). This OTM €60 call option would be worth zero, as the stock in this example cannot rise above €60.

ITM options trade less than OTM options as they are more expensive

An in-the-money (ITM) option is one which has a strike far away from spot and positive intrinsic value. Due to the positive intrinsic value ITM options are relatively expensive, hence tend to trade less than their cheaper OTM counterparts.

1.2: OPTION TRADING IN PRACTICE

Using options to invest has many advantages over investing in cash equity. Options provide leverage and an ability to take a view on volatility as well as equity direction. However, investing in options is more complicated than investing in equity, as a strike and expiry need to be chosen. This section explains hidden risks, (e.g. dividends) and other practical aspects of option trading such as how to choose the strike, and the difference between delta and the probability an option ends up ITM.

CHOOSING EXPIRY IS THE MOST DIFFICULT DECISION

The biggest difference between using options and cash equities (or delta 1 products) to gain equity exposure is the fact a suitable expiry has to be chosen. Determining if an equity is cheap or expensive is often easier than determining the driver and timing of the likely increase / decrease.

Choosing a far dated expiry gives most opportunity for the expected correction, however far dated options are very expensive. Conversely if a cheaper near dated expiry is chosen, there is little time for the anticipated movement to occur. Usually key dates such as quarterly reporting or elections help determine a suitable expiry.

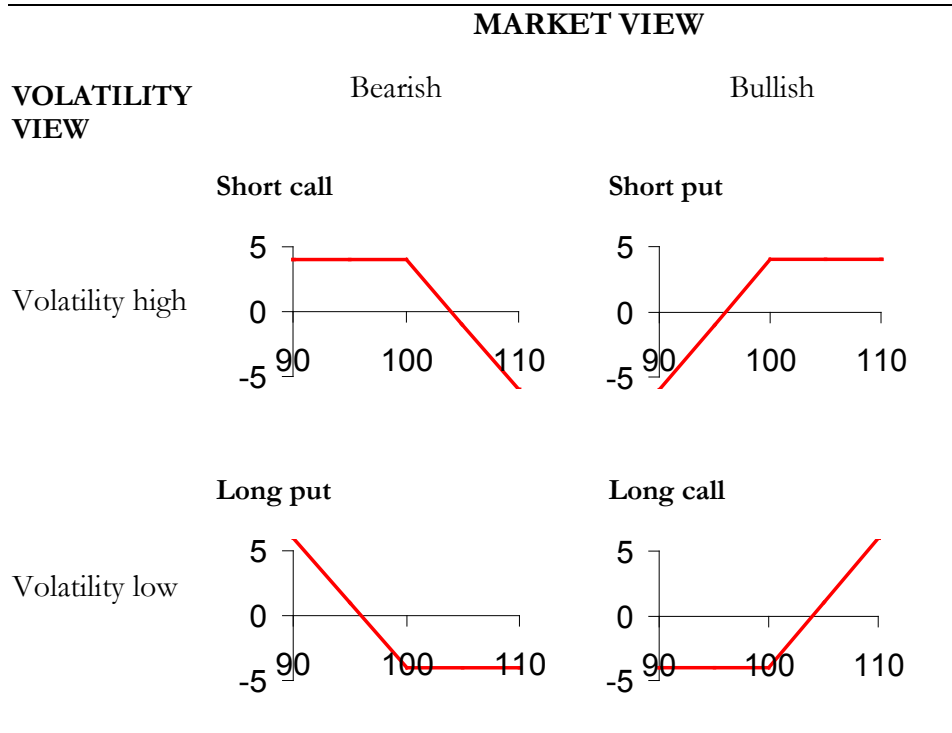
Expiry choice enforces investor discipline

Having to choose an expiry can be seen as a disadvantage of option trading, but some investors see it as an advantage as it enforces investor discipline. The process of choosing an expiry focuses attention on the likely dates a stock will converge with a forecast target price.

If the stock performs as expected the process of expiration forces the profits on the option position to be taken, and ensures a position is not held longer than it should be. Additionally if the anticipated return has not occurred by the expected date, the position expires worthless and forces the investor to make a decision if another position should be initiated. Using options to gain equity exposure therefore prevents “inertia” in a portfolio.

Both an equity and volatility view is needed to trade options

Option trading allows a view on equity, and volatility markets to be taken. If implied volatility is seen to be expensive then a short volatility strategy is best (short put for a bullish strategy, short call for a bearish strategy). However if implied volatility is seen to be cheap then a long volatility strategy is best (long call for a bullish strategy, long put for a bearish strategy). The appropriate strategy for a one leg option trade is shown in Figure 1 below. For multiple leg strategies see the section [*1.6 Option Structures Trading*](#).

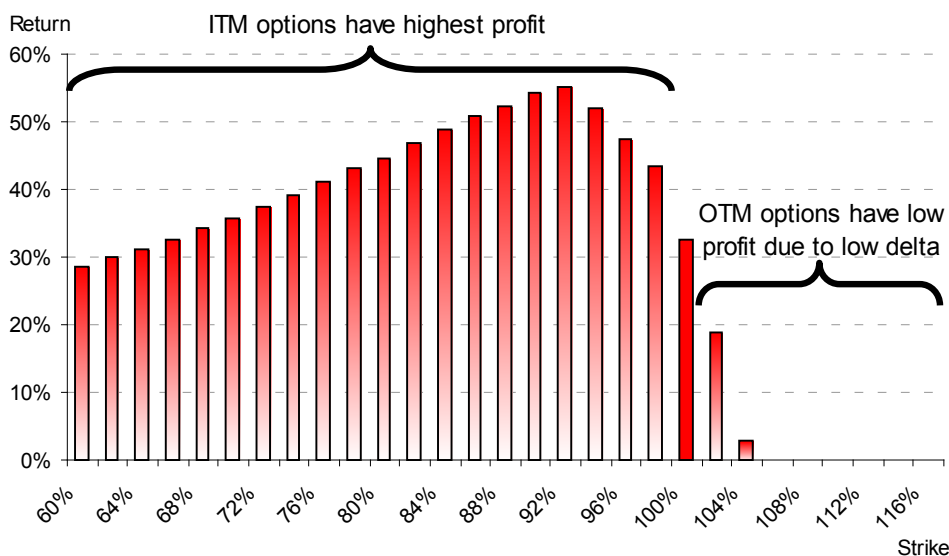
Figure 1. Option Strategy for Different Market and Volatility Views

Long vol strategies should have expiry just after key date

Typically if a key date is likely to be volatile then a long volatility strategy (long call or long put) should have an expiry just after this date. Conversely a short volatility strategy (short call or short put) should have an expiry just before the key date. For investors who wish to trade the implied jump of a key date, details of how to trade this implied jump is dealt with in the section [*6.4: Trading Earnings Announcements/Jumps*](#).

CHOOSING STRIKE OF STRATEGY IS NOT TRIVIAL

While choosing the strike of a strategy is not as difficult as choosing the expiry, it is not trivial. Investors could choose ATM to benefit from greatest liquidity. Alternatively, they could look at the highest expected return (option payout less the premium paid, as a percentage of the premium paid).

Figure 2. Profit of 12 Month Options if Markets Rise 10% by Expiry

ITM options have highest return for “normal” market moves

While choosing a cheap OTM option might be thought of as giving the highest return, the Figure below shows that, in fact, the highest returns come from in-the-money (ITM) options (ITM options have a strike far away from spot and have intrinsic value). This is because an ITM option has a high delta (sensitivity to equity price); hence, if an investor is relatively confident of a specific return, an ITM option has the highest return for relatively “normal” market moves (as trading an ITM option is similar to trading a forward).

Forwards are better than options for pure directional plays

A forward is a contract that obliges the investor to buy a security on a certain expiry date at a certain strike price. A forward has a delta of 100%. An ITM call option has many similarities with being long a forward, as it has a relatively small time value (compared to ATM) and a delta close to 100%. While the intrinsic value does make the option more expensive, this intrinsic value is returned at expiry. However, for an ATM option, the time value purchased is deducted from the returns. For pure directional plays, forwards (or futures, their listed equivalent) are more profitable than options. The advantage of options is in offering convexity: if markets move against the investor the only loss is the premium paid, whereas a forward has a virtually unlimited loss.

OTM options have highest return for “abnormal” moves

Only if the expected return is relatively high (or abnormal) do ATM or OTM options have the highest return. This is because for exceptional returns their low cost and high leverage more than compensates for their lower delta.

LIQUIDITY CAN BE A FACTOR IN CHOOSING STRIKE

If an underlying is relatively illiquid, or if the size of the trade is large, an investor should take into account the liquidity of the maturity and strike of the option. Typically, OTM options are more liquid than ITM options as ITM options tie up a lot of capital. This means that for strikes less than spot, puts are more liquid than calls and vice versa.

Low strike puts are usually more liquid than high strike calls

We note that as low-strike puts have a higher implied than high-strike calls, their value is greater and, hence, traders are more willing to use them. Low strike put options are therefore usually more liquid than high-strike call options. In addition, demand for protection lifts liquidity for low strikes compared with high strikes.

Single stock liquidity is limited for maturities up to two years

For single stock options, liquidity starts to fade after one year and options rarely trade over two years. For indices, longer maturities are liquid, partly due to the demand for long-dated hedges and their use in structured products. While structured products can have a maturity of five to ten years, investors typically lose interest after a few years and sell the product back. The hedging of a structured product, therefore, tends to be focused on more liquid maturities of around three years.

Hedge funds and structured product flow can overlap

Hedge funds tend to focus around the one-year maturity, with two to three years being the longest maturity they will consider. The two-to-three year maturity is where there is greatest overlap between hedge funds and structured desks.

DELTA MEASURES DIVIDEND RISK AND EQUITY RISK

The delta of the option is the amount of equity market exposure an option has. As a stock price falls by the dividend amount on its ex-date, delta is equal to the exposure to dividends that go ex before expiry. The dividend risk is equal to the negative of the delta. For example, if you have a call of positive delta, if (expected or actual) dividends rise, the call is worth less (as the stock falls by the dividend amount).

If a dividend is substantial, it could be in an investor's interest to exercise early. For more details, see the section [*1.3 Maintenance of Option Positions*](#).

DELTA IS NOT THE PROBABILITY OPTION EXPIRES ITM

A digital call option is an option that pays 100% if spot expires above the strike price (a digital put pays 100% if spot is below the strike price). The probability of such an option expiring ITM is equal to its delta, as the payoff only depends on it being ITM or not (the size of the payment does not change with how much ITM spot is). For a vanilla option this is not the case; hence, there is a difference between the delta and the probability of being ITM. This difference is typically small unless the maturity of the option is very long.

Delta takes into account the amount an option can be ITM

While a call can have an infinite payoff, a put's maximum value is the strike (as spot cannot go below zero). The delta hedge for the option has to take this into account, so a call delta must be greater than the probability of being ITM. Similarly, the absolute value (as put deltas are negative) of the put delta must be less than the probability of expiring ITM. A more mathematical explanation (for European options) is given below:

Call delta $>$ Probability call ends up ITM

Abs (Put delta) $<$ Probability put ends up ITM

Mathematical proof option delta is different from probability of being ITM at expiry

Call delta $= N(d_1)$ Put delta $= N(d_1) - 1$

Call probability ITM $= N(d_2)$ Put probability ITM $= 1 - N(d_2)$

where:

Definition of d_1 is the standard Black-Scholes formula for d_1 . For more details, see the section [4.7 Black-Scholes Formula](#).

$d_2 = d_1 - \sigma\sqrt{T}$

σ = implied volatility

T = time to expiry

$N(z)$ = cumulative normal distribution

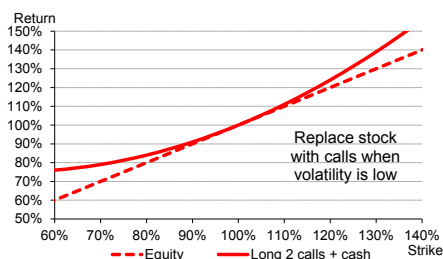
As d_2 is less than d_1 (see above) and $N(z)$ is a monotonically increasing function, this means that $N(d_2)$ is less than $N(d_1)$. Hence, the probability of a call being in the money = $N(d_2)$ is less than the delta = $N(d_1)$. As the delta of a put = delta of call – 1, and the sum of call and put being ITM = 1, the above results for a put must be true as well.

The difference between delta and probability being ITM at expiry is greatest for long-dated options with high volatility (as the difference between d_1 and d_2 is greatest for them).

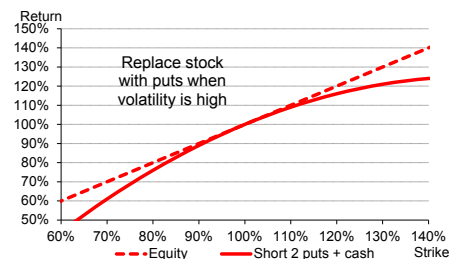
STOCK REPLACING WITH LONG CALL OR SHORT PUT

As a stock has a delta of 100%, the identical exposure to the equity market can be obtained by purchasing calls (or selling puts) whose total delta is 100%. For example, one stock could be replaced by two 50% delta calls, or by going short two -50% delta puts. Such a strategy can benefit from buying (or selling) expensive implied volatility. There can also be benefits from a tax perspective and, potentially, from any embedded borrow cost in the price of options (price of positive delta option strategies is improved by borrow cost). As the proceeds from selling the stock are typically greater than the cost of the calls (or margin requirement of the short put), the difference can be invested to earn interest.

Figure 3. Stock Replacing with Calls



Stock Replacing with Puts



Stock replacing via calls benefits from convexity

As a call option is convex, this means that the delta increases as spot increases and vice versa. If a long position in the underlying is sold and replaced with calls of equal delta, then if markets rise the delta increases and the calls make more money than the long position would have. Similarly, if markets fall the delta decreases and the losses are reduced. This can be seen in Figure 3 above as the portfolio of cash (proceeds from sale of the underlying) and call options is always above the long underlying profile. The downside of using calls is that the position will give a worse profile than the original long position if the underlying does not move much (as call options will fall each day by the theta if spot remains unchanged). Using call options is best when implied volatility is cheap and the investor expects the stock to move by more than currently implied.

Put underwriting benefits from selling expensive implied

Typically the implied volatility of options trades slightly above the expected realised volatility of the underlying over the life of the option (due to a mismatch between supply and demand). Stock replacement via put selling therefore benefits from selling (on average) expensive volatility. Selling a naked put is known as put underwriting, as the investor has effectively underwritten the stock (in the same way investment banks underwrite a rights issue).

Put underwriting pays investors for work that otherwise might be wasted

The strike of put underwriting should be chosen at the highest level at which the investor would wish to purchase the stock, which allows an investor to earn a premium from taking this view (whereas normally the work done to establish an attractive entry point would be wasted if the stock did not fall to that level).

Asset allocators use put underwriting to rebalance portfolios

This strategy has been used significantly recently by asset allocators who are underweight equities and are waiting for a better entry point to re-enter the equity market (earning the premium provides a buffer should equities rally). If an investor does not wish to own the stock and only wants to earn the premium, then an OTM strike should be chosen at a support level that is likely to remain firm.

Put underwriting benefits from selling skew

Put underwriting gives a similar profile to a long stock, short call profile, otherwise known as call overwriting. One difference between call overwriting and put underwriting is that if OTM options are used, then put underwriting benefits from selling skew (which is normally overpriced). For more details on the benefits of selling volatility, see the section [1.4 Call Overwriting](#).

STOCK REPLACEMENT ALTERS DIVIDEND EXPOSURE

It is important to note that the dividend exposure is not the same, as only the owner of a stock receives dividends. While the option owner does not benefit directly, the expected dividend will be used to price the option fairly (hence investors only suffer/benefit if dividends are different from expectations).

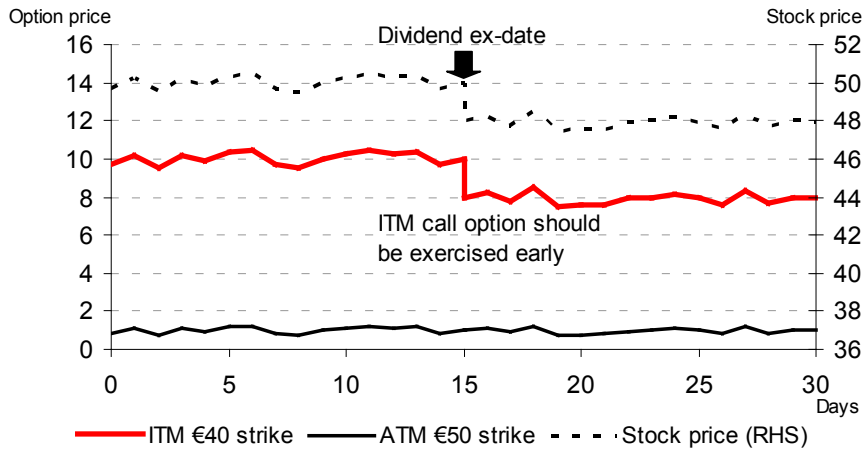
1.3: MAINTENANCE OF OPTION POSITIONS

During the life of an American option many events can occur in which it might be preferable to own the underlying shares (rather than the option) and exercise early. In addition to dividends, an investor might want the voting rights, or alternatively might want to sell the option to purchase another option (rolling the option). We investigate these life cycle events and explain when it is in an investor's interest to exercise, or roll, an option before expiry.

CONVERTING OPTIONS EARLY IS RARE

Options on indices are usually European, which means they can only be exercised at maturity. The inclusion of automatic exercise, and the fact it is impossible to exercise before maturity, means European options require only minimal maintenance. Single stock options, however, are typically American (apart from emerging market underlyings). While American options are rarely exercised early, there are circumstances when it is in an investor's interest to exercise an ITM option early. For both calls and puts the correct decision for early exercise depends on the net benefit of doing so (ie, the difference between earning the interest on the strike and net present value of dividends) versus the time value of the option.

- **Calls should be exercised just before the ex-date of a large unadjusted dividend.** In order to exercise a call, the strike price needs to be paid. The interest on this strike price normally makes it unattractive to exercise early. However, if there is a large unadjusted dividend that goes ex before expiry, it might be in an investor's interest to exercise an ITM option early (see Figure 4 above). In this case, the time value should be less than the dividend NPV (net present value) less total interest r ($=e^{r_{fr} \times T} - 1$) earned on the strike price K . In order to maximise 'dividend NPV - Kr ', it is best to exercise just before an ex-date (as this maximises 'dividend NPV' and minimises the total interest r).
- **Puts should be exercised early (preferably just after ex-date) if interest rates are high.** If interest rates are high, then the interest r from putting the stock back at a high strike price K (less dividend NPV) might be greater than the time value. In this case, a put should be exercised early. In order to maximise ' Kr - dividend NPV', a put should preferably be exercised just after an ex-date.

Figure 4. Price of ITM and ATM Call Option with Stock Price over Div Ex-Date

Calls should be exercised early if there is a large dividend

The payout profile of a long call is similar to the payout of a long stock + long put of the same strike. As only ITM options should be exercised and as the strike of an ITM call means the put of the same strike is OTM, we shall use this relationship to calculate when an option should be exercised early.

An American call should only be exercised if it is in an investor's interest to exercise the option and buy a European put of the same strike (a European put of same strike will have the same time value as a European call if intrinsic value is assumed to be the forward).

- **Choice A:** Do not exercise. In this case there is no benefit or cost.
- **Choice B:** Borrow strike K at interest r ($=e^{rfr \times T}-1$) in order to exercise the American call. The called stock will earn the dividend NPV and the position has to be hedged with the purchase of a European put (of cost equal to the time value of a European call).

An investor will only exercise early if choice B > choice A.

- ⇒ $-Kr + \text{dividend NPV} - \text{time value} > 0$
- ⇒ $\text{dividend NPV} - Kr > \text{time value for American call to be exercised}$

Puts should only be exercised if interest earned (less dividends) exceeds time value

For puts, it is simplest to assume an investor is long stock and long an American put. This payout is similar to a long call of the same strike. An American put should only be exercised against the long stock in the same portfolio if it is in an investor's interest to exercise the option and buy a European call of the same strike.

- **Choice A:** Do not exercise. In this case the portfolio of long stock and long put benefits from the dividend NPV.
- **Choice B:** Exercise put against long stock, receiving strike K , which can earn interest r ($=e^{rf \times T} - 1$). The position has to be hedged with the purchase of a European call (of cost equal to the time value of a European put).

An investor will only exercise early if choice B > choice A

$$\Rightarrow Kr - \text{time value} > \text{dividend NPV}$$

$$\Rightarrow Kr - \text{dividend NPV} > \text{time value for American put to be exercised}$$

Selling ITM options that should be exercised early can be profitable

There have been occasions when traders deliberately sell ITM options that should be exercised early, hoping that some investors will forget. Even if the original counterparty is aware of this fact, exchanges randomly assign the counterparty to exercised options. As it is unlikely that 100% of investors will realise in time, such a strategy can be profitable.

ITM OPTIONS USUALLY EXERCISED AUTOMATICALLY

In order to prevent situations where an investor might suffer a loss if they do not give notice to exercise an ITM option in time, most exchanges have some form of automatic exercise. If an investor (for whatever reason) does not want the option to be automatically exercised, he must give instructions to that effect. The hurdle for automatic exercise is usually above ATM in order to account for any trading fees that might be incurred in selling the underlying post exercise.

Eurex automatic exercise has a higher hurdle than CBOE

For the CBOE, options are automatically exercised if they are US\$0.01 or more ITM (reduced in June 2008 from US\$0.05 or more), which is in line with Euronext-Liffe rules of a €0.01 or GBP0.01 minimum ITM hurdle. Eurex has a higher automatic hurdle, as a contract price has to be ITM by 99.99 or more (eg, for a euro-denominated stock with a contract size of 100 shares this means it needs to be at €0.9999 or more). Eurex does allow an investor to specify an automatic exercise level lower than the automatic hurdle, or a percentage of exercise price up to 9.99%.

CORPORATE ACTIONS CAN ADJUST STRIKE

While options do not adjust for ordinary dividends¹, they do adjust for special dividends. Different exchanges have different definitions of what is a special dividend, but typically it is considered special if it is declared as a special dividend, or is larger than a certain threshold (eg, 10% of the stock price). In addition, options are adjusted in the event of a corporate action, for example, a stock split or rights issue.

Equities and indices can treat bonus share issues differently

Options on equities and indices can treat bonus share issues differently. A stock dividend in lieu of an ordinary dividend is considered an ordinary dividend for options on an equity (hence is not adjusted) but is normally adjusted by the index provider.

Adjustment negates impact of dividend or corporate action

For both special dividends and corporate actions, the adjustment negates the impact of the event (principal of unchanged contract values), so the theoretical price of the options should be able to ignore the event. As the strike post adjustment will be non-standard, typically exchanges create a new set of options with the normal strikes. While older options can still trade, liquidity generally passes to the new standard strike options (particularly for longer maturities which do not have much open interest).

M&A AND SPINOFFS CAN CAUSE PROBLEMS

If a company spins off a subsidiary and gives shareholders shares in the new company, the underlying for the option turns into a basket of the original equity and the spun-off company. New options going into the original company are usually created, and the liquidity of the options into the basket is likely to fade. For a company that is taken over, the existing options in that company will convert into whatever shareholders were offered. If the acquisition was for stock, then the options convert into shares, but if the offer is partly in cash, then options can lose a lot of value as the volatility of cash is zero.

OPTIONS OFTEN ROLLED BEFORE EXPIRY

The time value of an option decays quicker for short-dated options than for far-dated options. To reduce the effect of time decay, investors often roll before expiry. For example, an investor could buy a one-year option and roll it after six months to a new one-year option.

¹ Some option markets adjust for all dividends.

1.4: CALL OVERWRITING

For a directional investor who owns a stock (or index), call overwriting by selling an OTM call is one of the most popular methods of yield enhancement. Historically, call overwriting (otherwise known as buy-write, as the stock is bought but a call is written against it) has been a profitable strategy due to implied volatility usually being overpriced. However, call overwriting does underperform in volatile, strongly rising equity markets. Overwriting with the shortest maturity is best, and the strike should be slightly OTM for optimum returns.

IMPLIED VOLATILITY IS USUALLY OVERPRICED

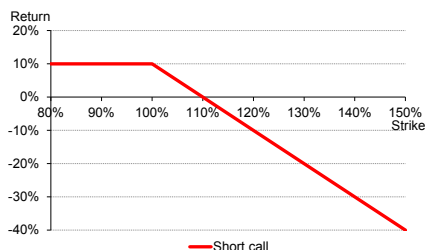
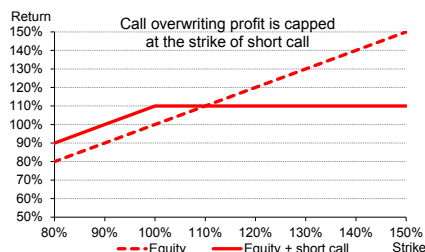
The implied volatility of options is on average 1-2pts above the volatility realised over the life of the option. This 'implied volatility premium' is usually greater for indices than for single stocks. As we can see no reason why these imbalances will fade, we expect call overwriting to continue to outperform on average. The key imbalances are:

- **Option buying for protection.** In the same way that no one buys car insurance because they think it is a good investment, investors are happy buying expensive protection to protect against downside risks.
- **Unwillingness to sell low premium options causes market makers to raise their prices** (selling low premium options, like selling lottery tickets, has to be done on a large scale to be attractive).
- **High gamma of near-dated options has a gap risk premium** (risk of stock jumping, either intraday or between closing and opening prices).
- **Index implieds lifted by structured products.** Structured products are often based on an index, and can offer downside protection. This lifts index implied relative to single stock implied. Also protection is usually bought on an index to protect against macros risks. It is rare to protect a single stock position (if an investor is worried about downside in a stock they usually do not buy it to begin with).

BUY-WRITE BENEFITS FROM SELLING EXPENSIVE VOL

Short-dated implied volatility has historically been overpriced² due to the above supply and demand imbalances. In order to profit from this characteristic, a long investor can sell a call against a long position in the underlying of the option. Should the underlying perform well and the call be exercised, the underlying can be used to satisfy the exercise of the call. As equities should be assumed to have, on average, a positive return, it is best to overwrite with a slightly OTM option to reduce the probability of the option sold expiring ITM.

² We note that implied volatility is not necessarily as overpriced as would first appear. For more detail, see the section *3.1 Implied Volatility Should Be Above Realized Volatility*.

Figure 5. Short Call**Call Overwriting (or Buy Write)**

Call overwriting is a useful way to gain yield in flat markets

If markets are range trading, or are approaching a technical resistance level, then selling a call at the top of the range (or resistance level) is a useful way of gaining yield. Such a strategy can be a useful tactical way of earning income on a core strategic portfolio, or potentially could be used as part of an exit strategy for a given target price.

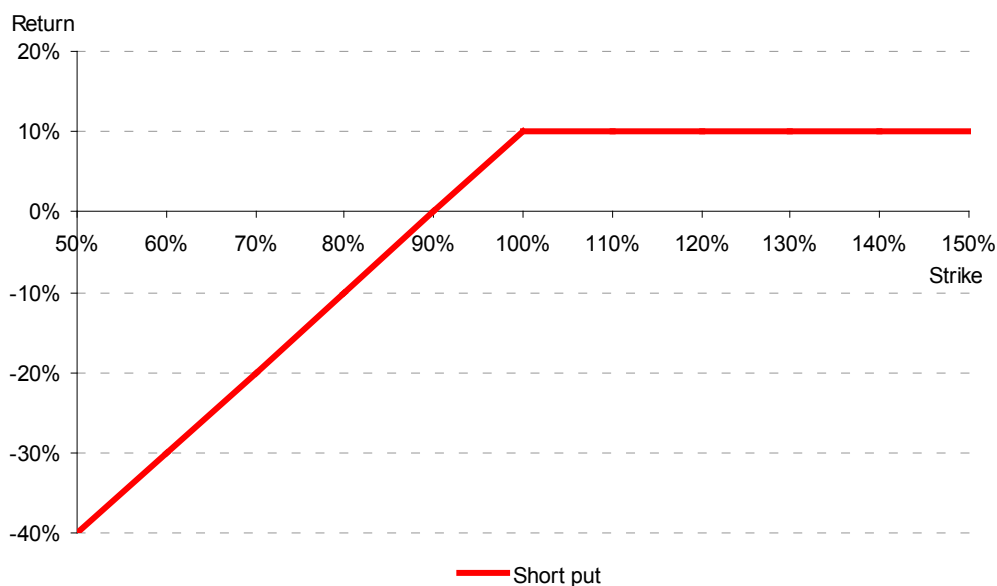
Selling at target price enforces disciplined investing

If a stock reaches the desired target price, there is the temptation to continue to own the strong performer. Over time a portfolio can run the risk of being a collection of stocks that had previously been undervalued, but are now at fair value. To prevent this inertia diluting the performance of a fund, some fund managers prefer to call overwrite at their target price to enforce disciplined investing, (as the stock will be called away when it reaches the target). As there are typically more Buy recommendations than Sell recommendations, call overwriting can ensure a better balance between the purchase and (called away) sale of stocks.

PUT UNDERWRITING HAS SIMILAR PROFILE

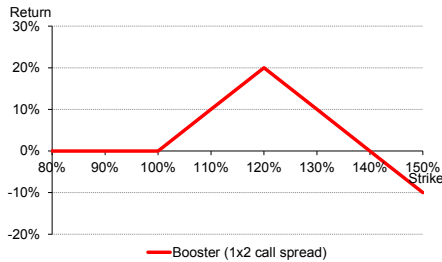
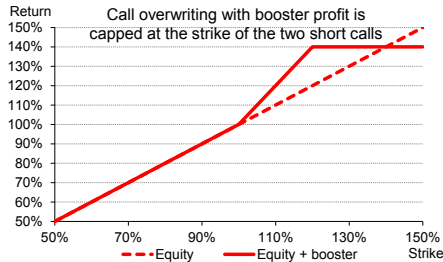
Figure 5 above shows the profiles of a short call and of a long equity with an overwritten call. The resulting profile of call overwriting is similar to that of a short put (Figure 6 below); hence, call overwriting could be considered similar to stock replacement with a short put (or put underwriting). Both call overwriting and put underwriting attempt to profit from the fact that implied volatility, on average, tends to be overpriced. While selling a naked put is seen as risky, due to the near infinite losses should stock prices fall, selling a call against a long equity position is seen as less risky (as the equity can be delivered against the exercise of the call).

Figure 6. Put Underwriting



1×2 call spreads are useful when a bounce-back is expected

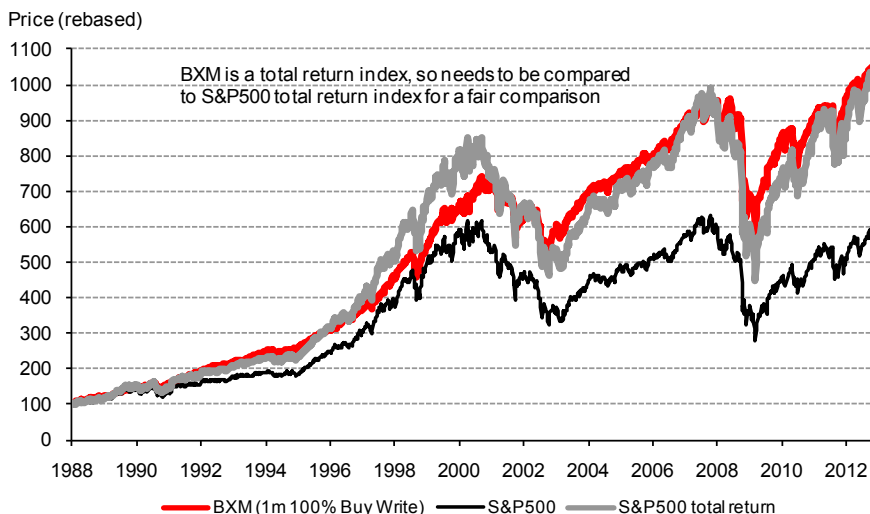
If a near zero cost 1×2 call spread (long 1×ATM call, short 2×OTM calls) is overlaid on a long stock position, the resulting position offers the investor twice the return for equity increases up to the short upper strike. For very high returns the payout is capped, in a similar way as for call overwriting. Such positioning is useful when there has been a sharp drop in the markets and a limited bounce back to earlier levels is anticipated. The level of the bounce back should be in line with or below the short upper strike. Typically, short maturities are best (less than three months) as the profile of a 1×2 call spread is similar to a short call for longer maturities.

Figure 7. Booster (1x2 Call Spread)**Call Overwriting with Booster**

CALL OVERWRITING IS BEST DONE ON AN INDEX

Many investors call overwrite on single stocks. However, single-stock implied volatility trades more in line with realised volatility than index implieds. The reason why index implieds are more overpriced than single-stock implieds is due to the demand from hedgers and structured product sellers. Call overwriting at the index level also reduces trading costs (due to the narrower bid-offer spread).

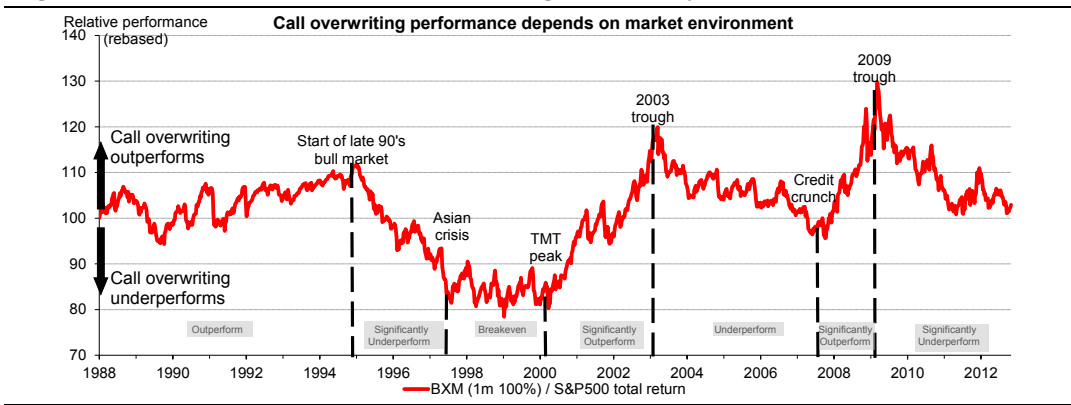
The CBOE has created a one-month call overwriting index on the S&P500 (BXM index), which is the longest call overwriting time series available. It is important to note that the BXM is a total return index; hence, it needs to be compared to the S&P500 total return index (SPXT Bloomberg code) not the S&P500 price return (SPX Bloomberg code). As can be seen in Figure 8 below, comparing the BXM index to the S&P500 price return index artificially flattens the performance of call overwriting.

Figure 8. S&P500 and S&P500 1M ATM Call Overwriting Index (BXM)

Call overwriting performance varies according to conditions

On average, ATM index call overwriting has been a profitable strategy. However, there have been periods of time when it has been unprofitable. The best way to examine the returns under different market conditions is to divide the BXM index by the total return S&P500 index (as the BXM is a total return index).

Figure 9. S&P500 1M ATM Call Overwriting Divided by S&P500 Total Return



Overwriting underperforms in bull markets with low volatility

Since the BXM index was created, there have been seven distinct periods (see Figure 9 above), each with different equity and volatility market conditions. Of the seven periods, the two in which returns for call overwriting are negative are the bull markets of the mid-1990s and middle of the last decade. These were markets with very low volatility, causing the short call option sold to earn insufficient premium to compensate for the option being ITM.

It is important to note that call overwriting can outperform in slowly rising markets, as the premium earned is in excess of the amount the option ends up ITM. This was the case for the BXM between 1986 and the mid-1990s. It is difficult to identify these periods in advance as there is a very low correlation between BXM outperformance and the earlier historical volatility.

LOWER EXPOSURE TO EQUITY RISK PREMIUM

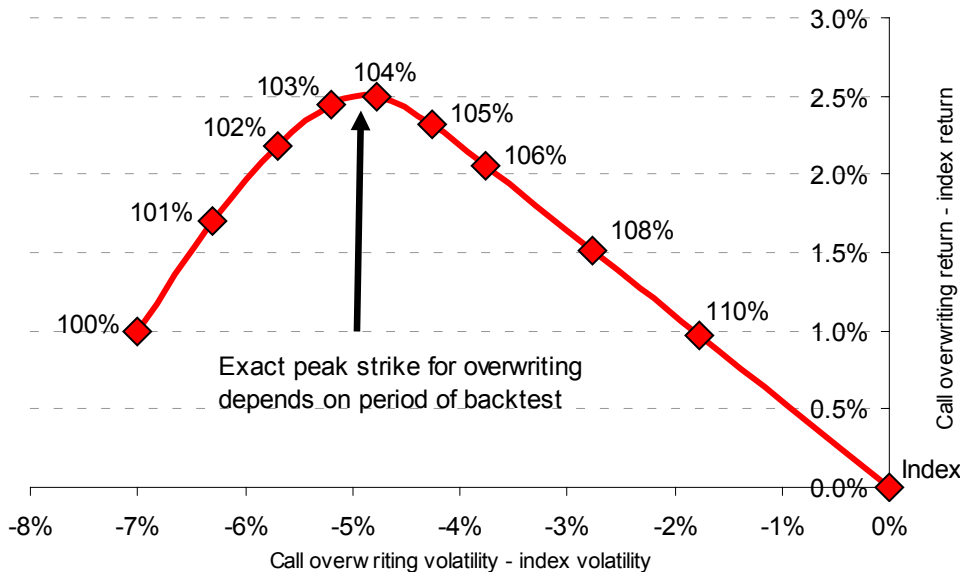
We note that while profits should be earned from selling an expensive call, the delta (or equity sensitivity) of the long underlying short call portfolio is significantly less than 100% (even if the premium from the short call is reinvested into the strategy). Assuming that equities are expected to earn more than the risk free rate (ie, have a positive equity risk premium), this lower delta can mean more money is lost by having a less equity-sensitive portfolio than is gained by selling expensive volatility. On average, call overwriting appears

to be a successful strategy, and its success has meant that it is one of the most popular uses of trading options.

OVERWRITING WITH NEAR-DATED OPTIONS IS BEST

Near-dated options have the highest theta, so an investor earns the greatest carry from call overwriting with short-dated options. It is possible to overwrite with 12 one-month options in a year, as opposed to four three-month options or one 12-month option. While overwriting with the shortest maturity possible has the highest returns on average, the strategy does have potentially higher risk. If a market rises one month, then retreats back to its original value by the end of the quarter, a one-month call overwriting strategy will have suffered a loss on the first call sold but a three-month overwriting strategy will not have had a call expire ITM. However, overwriting with far-dated expiries is more likely to eliminate the equity risk premium the investor is trying to earn (as any outperformance above a certain level will be called away).

Figure 10. Call Overwriting SX5E with One-Month Calls of Different Strikes



BEST RETURNS WITH SLIGHTLY OTM OPTIONS

While overwriting with near-dated expiries is clearly superior to overwriting with far-dated expiries, the optimal choice of strike to overwrite with depends on the market environment. As equities are expected, on average, to post a positive return, overwriting should be done with slightly OTM options. However, if a period of time where equities had a negative return is chosen for a back-test, then a strike below 100% could show the highest return. Looking

at a period of time where the SX5E had a positive return shows that for one-month options a strike between 103%-104% is best (see Figure 10 above).

Typically call overwriting with c25% delta call options is best

For three-month options, the optimal strike is a higher 107%-108%, but the outperformance is approximately half as good as for one-month options. These optimal strikes for overwriting could be seen to be arguably high, as recently there have been instances of severe declines (TMT bubble bursting, Lehman bankruptcy), which were followed by significant price rises afterwards. For single-stock call overwriting, these strikes could seem to be low, as single stocks are more volatile. For this reason, many investors use the current level of volatility to determine the strike or choose a fixed delta option (eg, 25%).

OVERWRITING REDUCES VOLATILITY

While selling an option could be considered risky, the volatility of returns from overwriting a long equity position is reduced by overwriting. This is because the payout profile is capped for equity prices above the strike. An alternative way of looking at this is that the delta of the portfolio is reduced from 100% (solely invested in equity) to 100% less the delta of the call (c50% depending on strike). The reduced delta suppresses the volatility of the portfolio.

Risk reduction less impressive if Sortino ratios are used

We note that the low call overwriting volatility is due to the lack of volatility to the upside, as call overwriting has the same downside risk as a long position. For this reason, using the Sortino ratio (for more details, see the section *A11 Sortino Ratio* in the *Appendix*) is likely to be a fairer measure of call overwriting risk than standard deviation, as standard deviation is not a good measure of risk for skewed distributions. Sortino ratios show that the call overwriting downside risk is identical to a long position; hence, call overwriting should primarily be done to enhance returns and is not a viable strategy for risk reduction.

Optimal strike is similar for single stocks and indices

While this analysis is focused on the SX5E, the analysis can be used to guide single-stock call overwriting (although the strike could be adjusted higher by the single-stock implied divided by SX5E implied).

ENHANCED CALL OVERWRITING IS DIFFICULT

Enhanced call overwriting is the term given when call overwriting is only done opportunistically or the parameters (strike or expiry) are varied according to market conditions. On the index level, the returns from call overwriting are so high that enhanced call overwriting is difficult, as the opportunity cost from not always overwriting is too high. For single stocks, the returns for call overwriting are less impressive; hence, enhanced call overwriting could be more successful. An example of single-stock enhanced call overwriting is to only overwrite when an implied is high compared to peers in the same sector. We note that even with enhanced single-stock call overwriting, the wider bid-offer cost and smaller implied volatility premium to realised means returns can be lower than call overwriting at the index level.

Enhanced call overwriting returns likely to be arbitrated away

Should a systematic way to enhance call overwriting be viable, this method could be applied to volatility trading without needing an existing long position in the underlying. Given the presence of statistical arbitrage funds and high frequency traders, we believe it is unlikely that a simple automated enhanced call overwriting strategy on equity or volatility markets is likely to outperform vanilla call overwriting on an ongoing basis.

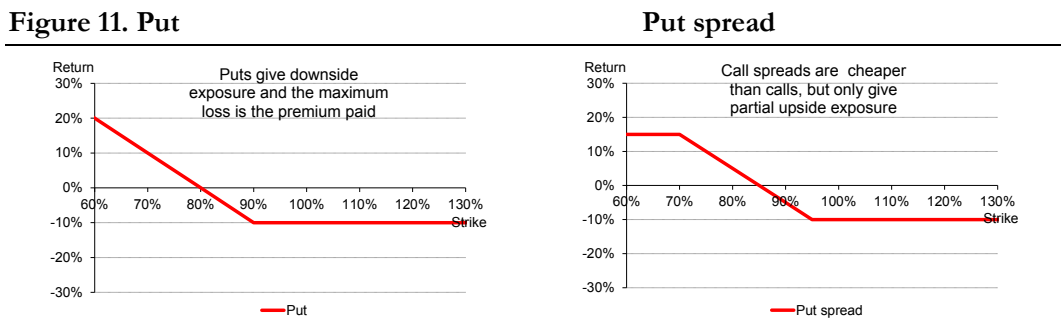
1.5: PROTECTION STRATEGIES USING OPTIONS

For both economic and regulatory reasons, one of the most popular uses of options is to provide protection against a long position in the underlying. The cost of buying protection through a put is lowest in calm, low-volatility markets, but in more turbulent markets the cost can be too high. In order to reduce the cost of buying protection in volatile markets (which is often when protection is in most demand), many investors sell an OTM put and/or an OTM call to lower the cost of the long put protection bought.

CHEAPEN PROTECTION BY SELLING OTM PUT & CALL

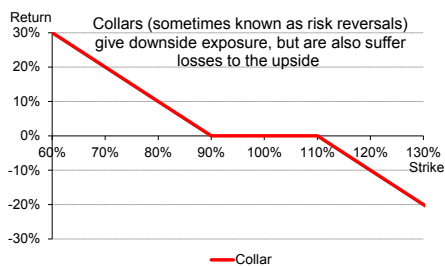
Buying a put against a long position gives complete and total protection for underlying moves below the strike (as the investor can simply put the long position back for the strike price following severe declines). The disadvantage of a put is the relatively high cost, as an investor is typically unwilling to pay more than 1%-2% for protection (as the cost of protection usually has to be made up through alpha to avoid underperforming if markets do not decline). The cost of the long put protection can be cheapened by selling an OTM put (turning the long put into a long put spread), by selling an OTM call (turning put protection into a collar), or both (resulting in a put spread vs call, or put spread collar). The strikes of the OTM puts and calls sold can be chosen to be in line with technical supports or resistance levels.

Figure 11. Put

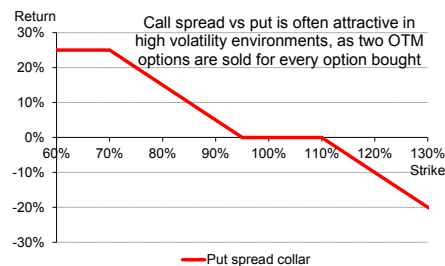


- **Puts give complete protection without capping performance.** As puts give such good protection, their cost is usually prohibitive unless the strike is low. For this reason, put protection is normally bought for strikes around 90%. Given that this protection will not kick in until there is a decline of 10% or more, puts offer the most cost-effective protection only during a severe crash (or if very short-term protection is required).
- **Put spreads only give partial protection but are cost effective.** While puts give complete protection, often only partial protection is necessary, in which case selling an OTM put against the long put (a put spread) can be an attractive protection strategy. The value of the put sold can be used to either cheapen the protection or lift the strike of the long put.
- **Collars can be zero cost as they give up some upside.** While investors appreciate the need for protection, the cost needs to be funded through reduced performance (or less alpha) or by giving up some upside. Selling an OTM call to fund a put (a collar) results in a cap on performance. However, if the strike of the call is set at a reasonable level, the capped return could still be attractive. The strike of the OTM call is often chosen to give the collar a zero cost. Collars can be a visually attractive low (or zero) cost method of protection as returns can be floored at the largest tolerable loss and capped at the target return. A collar is unique among protection strategies in not having significant volatility exposure, as its profile is similar to a short position in the underlying. Collars are, however, exposed to skew.
- **Put spread collars best when volatility is high, as two OTM options are sold.** Selling both an OTM put and OTM call against a long put (a put spread collar) is typically attractive when volatility is high, as this lifts the value of the two OTM options sold more than the long put bought. If equity markets are range bound, a put spread collar can also be an attractive form of protection. Put spread collars are normally structured to be near zero cost (just like a collar).

Figure 12. Collar



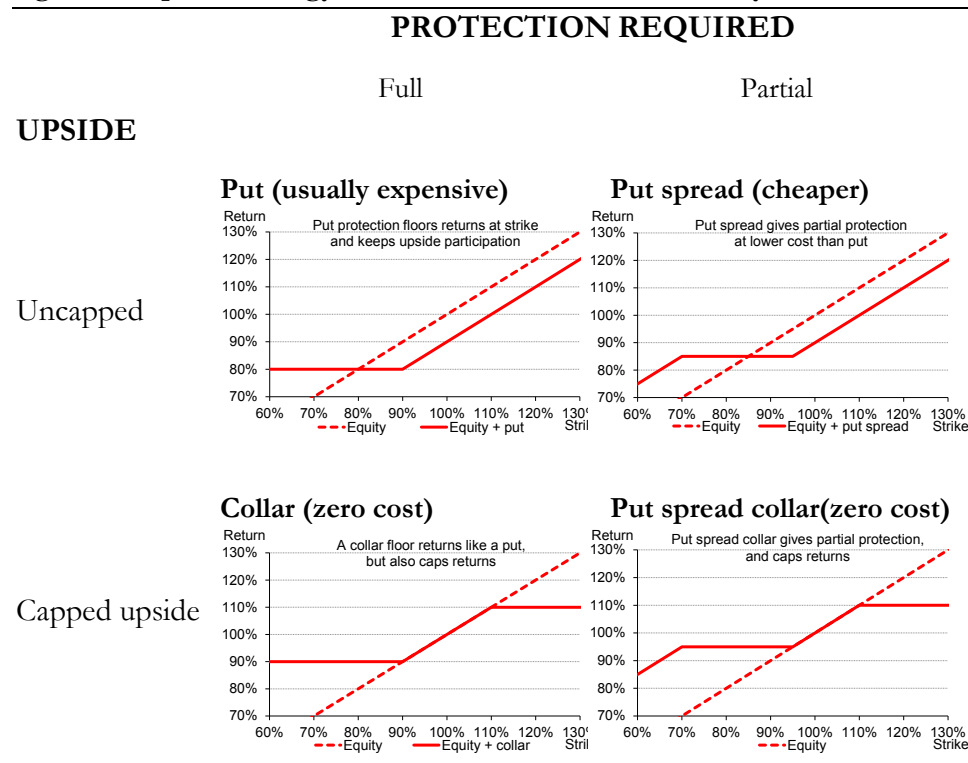
Put spread collar



Portfolio protection is usually done via indices to lower cost

While an equity investor will typically purchase individual stocks, if protection is bought then this is usually done at the index level. This is because the risk the investor wishes to hedge against is the general equity or macroeconomic risk. If a stock is seen as having excessive downside risk, it is usually sold rather than a put bought against it. An additional reason why index protection is more common than single stock protection is the fact that bid-offer spreads for single stocks are wider than for an index.

Figure 13. Option Strategy for Different Market and Volatility Views



Partial protection can give attractive risk reward profile

For six-month maturity options, the cost of a 90% put is typically in line with a 95%-85% put spread (except during periods of high volatility, when the cost of a put is usually more expensive). Put spreads often have an attractive risk-reward profile for protection of the same cost, as the strike of the long put can be higher than the long put of a put spread. Additionally, if an investor is concerned with outperforming peers, then a c10% outperformance given by a 95%-85% put spread should be sufficient to attract investors (there is little incremental competitive advantage in a greater outperformance).

Implied volatility is far more important than skew for put-spread pricing

A rule of thumb is that the value of the OTM put sold should be approximately one-third the value of the long put (if it were significantly less, the cost saving in moving from a put to a put spread would not compensate for giving up complete protection). While selling an OTM put against a near-ATM put does benefit from selling skew (as the implied volatility of the OTM put sold is higher than the volatility of the near ATM long put bought), the effect of skew on put spread pricing is not normally that significant (far more significant is the level of implied volatility).

Collars are more sensitive to skew than implied volatility

Selling a call against a long put suffers from buying skew. The effect of skew is greater for a collar than for a put spread, as skew affects both legs of the structure the same way (whereas the effect of skew on the long and short put of a put spread partly cancels). If skew was flat, the cost of a collar typically reduces by 1% of spot. The level of volatility for near-zero cost collars is not normally significant, as the long volatility of the put cancels the short volatility of the call.

Capping performance should only be used when a long rally is unlikely

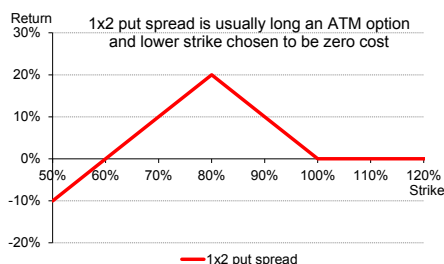
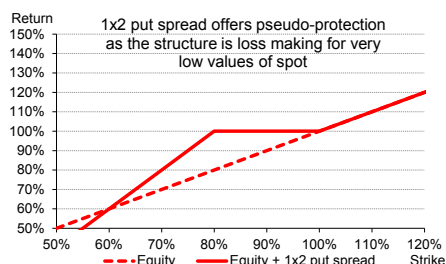
A collar or put spread collar caps the performance of the portfolio at the strike of the OTM call sold. They should only therefore be used when the likelihood of a strong, long-lasting rally (or significant bounce) is perceived to be relatively small.

Bullish investors could sell two puts against long put

If an investor is bullish on the equity market, then a protection strategy that caps performance is unsuitable. Additionally, as the likelihood of substantial declines is seen to be small, the cost of protection via a put or put spread is too high. In this scenario, a zero cost 1×2 put spread could be used as a pseudo-protection strategy. The long put is normally ATM, which means the portfolio is 100% protected against falls up to the lower strike, and gives partial protection below that until the breakeven. A loss is only suffered if the equity market falls below the breakeven.

1×2 put spreads only give pseudo-protection

We do not consider 1x2 put spreads to offer true protection, as during severe declines it will suffer a loss when the underlying portfolio is also heavily loss making. The payout of 1×2 put spreads for maturities of around three months or more is initially similar to a short put, so we consider it to be a bullish strategy. However, for the SX5E a roughly six-month zero-cost 1×2 put spread, whose upper strike is 95%, has historically had a breakeven below 80% and declines of more than 20% in six months are very rare. As 1×2 put spreads do not provide protection when you need it most, they could be seen as a separate long position rather than a protection strategy.

Figure 14. 1×2 Put Spread**1×2 Put Spread Pseudo-Protection**

PROTECTION MUST BE PAID FOR: QUESTION IS HOW?

If an investor seeks protection, the most important decision that has to be made is how to pay for it. The cost of protection can be paid for in one of three ways. Figure 15 below shows when this cost is suffered by the investor, and when the structure starts to provide protection against declines.

Premium. The simplest method of paying for protection is through premium. In this case, a put or put spread should be bought.

Loss of upside. If the likelihood of extremely high returns is small, or if a premium cannot be paid, then giving up upside via collars or put spread collars is the best way to pay for protection.

Potential losses on extreme downside. If an investor is willing to tolerate additional losses during extreme declines, then a 1×2 put spread can offer a zero cost way of buying protection against limited declines in the market.

Figure 15. Protection Strategy Comparison

Equity Performance	Put	Put Spread	Collar	Put Spread Collar	1×2 Put Spread
Bull markets (+10% or more)	Loss of premium	Loss of premium	Loss of upside	Loss of upside	–
Flat markets (±5%)	Loss of premium	Loss of premium	–	–	–
Moderate dip (c-10%)	Loss of premium	Protected	–	Protected	Protected
Correction (c-15%)	Protected	Protected	Protected	Protected	Protected
Bear market (c-20% or worse)	Protected	Partially protected	Protected	Partially protected	Severe loss

BEST STRATEGY DETERMINED BY VOLATILITY LEVEL

The level of volatility can determine the most suitable protection strategy an investor needs to decide how bullish and bearish they are on the equity and volatility markets. If volatility is low, then puts should be affordable enough to buy without offsetting the cost by selling an OTM option. For low to moderate levels of volatility, a put spread is likely to give the best protection that can be easily afforded. As a collar is similar to a short position with limited volatility exposure, it is most appropriate for a bearish investor during average periods of volatility (or if an investor does not have a strong view on volatility). Put spreads collars (or 1×2 put spreads) are most appropriate during high levels of volatility (as two options are sold for every option bought).

MATURITY DRIVEN BY DURATION OF LIKELY DECLINE

The choice of protection strategy is typically driven by an investor's view on equity and volatility markets. Similarly the choice of strikes is usually restricted by the premium an investor can afford. Maturity is potentially the area where there is most choice, and the final decision will be driven by an investor's belief in the severity and duration of any decline. If he wants protection against a sudden crash, a short-dated put is the most appropriate strategy. However, for a long drawn out bear market, a longer maturity is most appropriate.

Figure 16. Types of DAX Declines (of 10% or more) since 1960

Type	Average Decline	Decline Range	Average Duration	Duration Range
Crash	31%	19% to 39%	1 month	0 to 3 months
Correction	14%	10% to 22%	3 months	0 to 1 year
Bear market	44%	23% to 73%	2.5 years	1 to 5 years

Median maturity of protection bought is c4 months

The average choice of protection is c6 months, but this is skewed by a few long-dated hedges. The median maturity is c4 months. Protection can be bought for maturities of one week to over a year. Even if an investor has decided how long he needs protection, he can implement it via one far-dated option or multiple near-dated options. For example, one-year protection could be via a one-year put or via the purchase of a three-month put every three months (four puts over the course of a year). The typical cost of ATM puts for different maturities is given below.

Figure 17. Cost of ATM Put on SX5E

Cost	1 Month	2 Months	3 Months	6 Months	1 Year
Individual premium	2.3%	3.3%	4.0%	5.7%	8.0%
Rolling protection cost per year	27.7%	19.6%	16.0%	11.3%	8.0%

Short-dated puts offer greatest protection but highest cost

If equity markets fall 20% in the first three months of the year and recover to the earlier level by the end of the year, then a rolling three-month put strategy will have a positive payout in the first quarter but a one-year put will be worth nothing at expiry. While rolling near-dated puts will give greater protection than a long-dated put, the cost is higher (see Figure 17 above).

SHORT VOL AGAINST LONG PUT PERFORMS WELL

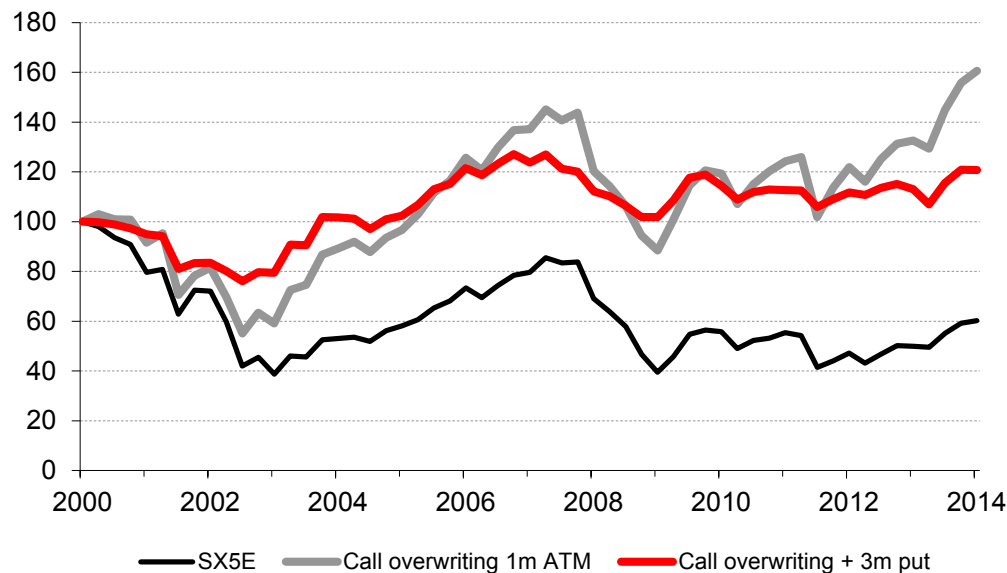
All protection strategies that combine a long and a short aim to offset the overpriced cost of protection by selling the same overpriced implied volatility for a different maturity and strike. Hence such strategies tend to back-test well as the overall exposure to expensive implied volatility is near zero. As the net cost of such strategies is near zero, while at the same time (usually) decreasing the volatility of the portfolio, their risk adjusted performance can be impressive. Their performance can often be further improved by selling near dated volatility against the long far dated protection.

MULTIPLE EXPIRY PROTECTION STRATEGIES

Typically, a protection strategy involving multiple options has the same maturity for all of the options. However, some investors choose a nearer maturity for the options they are short, as more premium can be earned selling a near-dated option multiple times (as near-dated options have higher theta). These strategies are most successful when term structure is inverted, as the volatility for the near-dated option sold is higher. Having a nearer maturity for the long put option and longer maturity for the short options makes less sense, as this increases the cost (assuming the nearer-dated put is rolled at expiry).

Calendar collar effectively overlays call overwriting on a long put position

If the maturity of the short call of a collar is closer than the maturity of the long put, then this is effectively the combination of a long put and call overwriting. For example, the cost of a three-month put can be recovered by selling one-month calls. This strategy outperforms in a downturn and also has a lower volatility (see Figure 18 below).

Figure 18. Performance of 3M Put vs 1M Call Overwriting

Selling a put against calendar collar creates either a calendar put spread collar, or a put vs strangle/straddle

If a put is sold against the position of a calendar collar, then the final position depends on the maturity of the short put. The short put can either have the maturity of the long put (creating a calendar put spread collar), or the maturity of the short call (creating a put vs strangle/straddle).

- **Calendar put spread collar.** If the maturity of the short put is identical to the long (far-dated) put, then the final position is a calendar put spread collar (i.e. far dated put spread funded by sale of short dated calls). The performance of a calendar put spread collar is similar to the calendar collar above.
- **Put vs strangle/straddle.** If the maturity of the short put is the same as the maturity of the short near-dated call, then this position funds the long far-dated put by selling near-dated volatility via a near dated strangle (or straddle if the strikes of the short put and short call are identical).

Short near dated variance swaps is an alternative to selling near dated strangle/straddle

For an investor who is able to trade OTC, a similar strategy involves long put and short near-dated variance swaps.

1.6: OPTION STRUCTURES TRADING

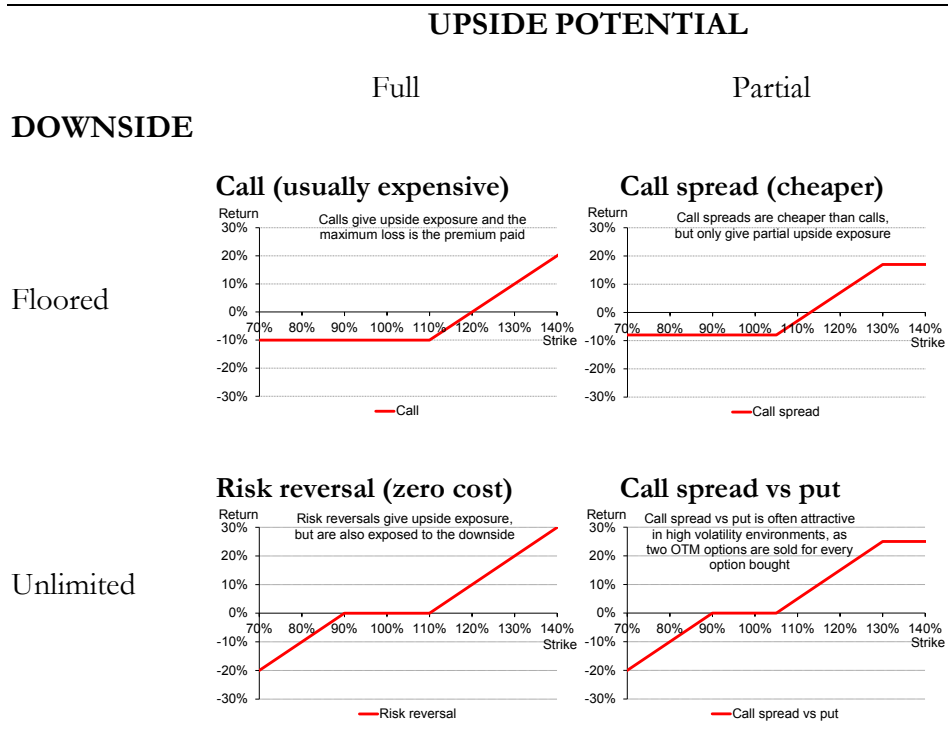
While a simple view on both volatility and equity market direction can be implemented via a long or short position in a call or put, a far wider set of payoffs is possible if two or three different options are used. We investigate strategies using option structures (or option combos) that can be used to meet different investor needs.

BULLISH COMBOS ARE REVERSE OF BEARISH

Using option structures to implement a bearish strategy has already been discussed in the section *1.5 Protection Strategies Using Options*. In the same way a long put protection can be cheapened by selling an OTM put against the put protection (to create a put spread giving only partial protection), a call can be cheapened by selling an OTM call (to create a call spread offering only partial upside). Similarly, the upside exposure of the call (or call spread) can be funded by put underwriting (just as put or put spread protection can be funded by call overwriting). The four option structures for bullish strategies are given below.

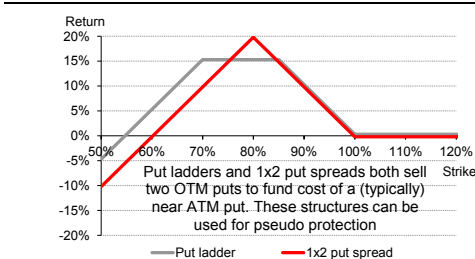
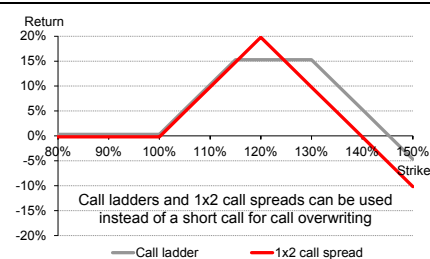
- **Calls give complete upside exposure and floored downside.** Calls are the ideal instrument for bullish investors as they offer full upside exposure and the maximum loss is only the premium paid. Unless the call is short dated or is purchased in a period of low volatility, the cost is likely to be high.
- **Call spreads give partial upside but are cheaper.** If an underlying is seen as unlikely to rise significantly, or if a call is too expensive, then selling an OTM call against the long call (to create a call spread) could be the best bullish strategy. The strike of the call sold could be chosen to be in line with a target price or technical resistance level. While the upside is limited to the difference between the two strikes, the cost of the strategy is normally one-third cheaper than the cost of the call.
- **Risk reversals (short put, long call of different strikes) benefit from selling skew.** If a long call position is funded by selling a put (to create a risk reversal), the volatility of the put sold is normally higher than the volatility of the call bought. The higher skew is, the larger this difference and the more attractive this strategy is. Similarly, if interest rates are low, then the lower the forward (which lifts the value of the put and decreases the value of the call) and the more attractive the strategy is. The profile of this risk reversal is similar to being long the underlying.
- **Call spread vs put is most attractive when volatility is high.** A long call can be funded by selling an OTM call and OTM put. This strategy is best when implied volatility is high, as two options are sold.

Figure 19. Upside Participation Strategies



LADDERS HAVE A SIMILAR PROFILE TO 1×2 SPREADS

With a 1×2 call or put spread, two OTM options of the same strike are sold against one (usually near ATM) long option of a different strike. A ladder has exactly the same structure, except the two short OTM options have a different strike. The different strikes of the two OTM options means that the maximum payout of a ladder is lower than for a 1x2, however this maximum payout is earned for a range of values, not a single value at expiry. Additionally the breakeven of a ladder is further away from spot, giving the comfort that markets have to move further for losses to be suffered when compared to a 1x2 call or put spread.

Figure 20. Put Ladders and 1×2**Call Ladders and 1×2**

STRADDLES AND STRANGLES ARE VERY SIMILAR

Using option structures allows a straddle (long call and put of same strike) or strangle (long call and put of different strikes) to be traded. These structures are long (or short) volatility, but do not have any exposure to the direction of the equity market.

STRUCTURES ALLOW WIDER RANGE OF PAYOUTS

Figure 21 below shows the most common structures that can be traded with up to three different options in relation to a view on equity and volatility markets. For simplicity, strangles and ladders are not shown, but they can be considered to be similar to straddles and 1×2 ratio spreads, respectively.

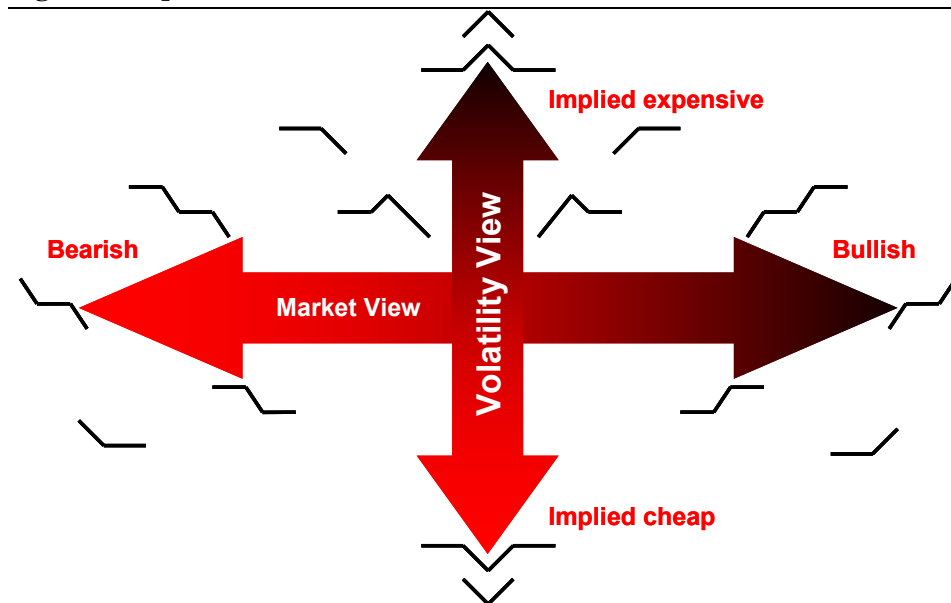
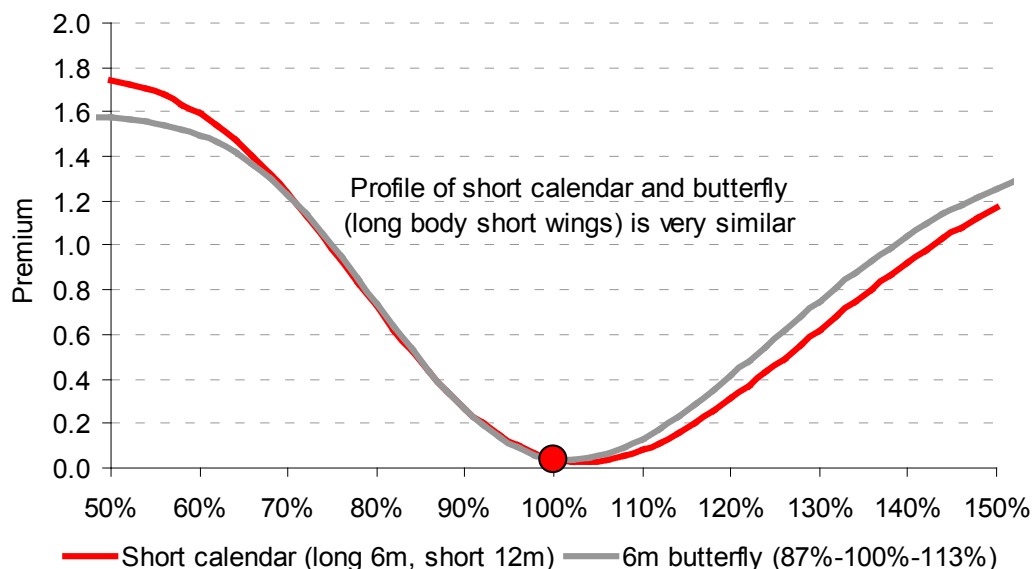
Figure 21. Option Structures

Figure 22. Theoretical Value of Butterfly and Short Calendar

1X1 CALENDAR TRADES ARE SIMILAR TO BUTTERFLY

Butterflies combine a short straddle with a long strangle, which allows a short volatility position to be taken with floored losses. We note the theoretical profile of a short calendar trade is similar to trading a butterfly. If an underlying does not have liquid OTM options, then a calendar can be used as a butterfly substitute (although this approach does involve term structure risk, which a butterfly does not have). A long calendar (short near-dated, long far-dated) is therefore short gamma (as the short near-dated option has more gamma than the far-dated option).